Information Sheet 2: Getting Started on the Modelling Server

EART22001 / EART60071

Aim. By the end of this session you should be able to log in to the modelling server, recognise whether a terminal is local or remote, create and edit a file, transfer it back to your own computer, and obtain practical code from GitHub.

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1 Find your course login details

Open Canvas and find the course server login details. You need a **course-specific username and password**; these are **not** your central University credentials.

For on-campus work the server address is:

```
130.88.55.57
```

Do not share your password or include it in screenshots. If Canvas announces a change to the server address, use the address on Canvas.

2 Open a terminal on your own computer

Windows: open PowerShell or Windows Terminal. Command Prompt also works.
macOS: press Command–Space, type *Terminal*, and press Return.

At this point the terminal is **local**: commands are running on your own computer.

3 Log in to the server with SSH

Type the following, replacing `YOUR_USERNAME` with the username from Canvas:

```
ssh YOUR_USERNAME@130.88.55.57
```

If SSH asks whether you trust the server the first time you connect, check that the address is exactly `130.88.55.57` and then answer `yes`.

Enter your course password. **Nothing appears while you type the password**; this is normal.

Once the prompt changes, you are on the **remote Linux server**. Try:

```
hostname
pwd
ls
```

Checkpoint: ask yourself which computer each of those commands ran on. The answer should be: the remote server.

4 Create a small text file with nano

At the remote prompt, type:

```
nano -l test.txt
```

The `-l` option displays line numbers. Type a short sentence, for example:

```
This file was created on the course modelling server.
```

Save with **Ctrl+O**, press Enter to accept the filename, then exit with **Ctrl+X**.

Check that the file exists:

```
ls -lh test.txt
```

5 Transfer the file to your own computer using SFTP

Leave the SSH window open. Open a **second terminal window** on your own computer. This second window is local.

It is easiest to choose the destination folder before starting SFTP.

Windows PowerShell:

```
cd $HOME\Downloads
sftp YOUR_USERNAME@130.88.55.57
```

macOS Terminal:

```
cd ~/Downloads
sftp YOUR_USERNAME@130.88.55.57
```

Enter the course password again. At the `sftp>` prompt, type:

```
lpwd
pwd
ls
get test.txt
exit
```

The two-directory rule in SFTP
`pwd` = directory on the **remote server**.
`lpwd` = directory on your **local computer**.
`get test.txt` copies the file from the remote `pwd` directory into the local `lpwd` directory.

Now reveal the local folder graphically.

Windows PowerShell:

```
explorer .
```

macOS Terminal:

```
open .
```

Open `test.txt` from Downloads and check that it contains the sentence you typed. If you cannot find it, do not repeatedly download it: reconnect with SFTP and use `lpwd` to find the local destination.

6 Practise a few safe Linux commands

Return to the **SSH window**. In the commands below, CAPITAL words are placeholders: replace them with an actual name. Do not type `DIR`, `FILE`, `OLDNAME` or `NEWNAME` literally.

Example to type	What it does
`mkdir practice`	make a directory called `practice`
`cd practice`	enter that directory
`pwd`	show exactly where you are
`cd ..`	go up one directory
`cp test.txt test_copy.txt`	make a copy of a file
`mv test_copy.txt test_old.txt`	rename or move a file
`rm -i test_old.txt`	remove a file, asking for confirmation first

Be careful with `rm`. Files deleted on the Linux server do not normally go to a recycle bin. For now, use `rm -i FILE`, check the filename, and answer the confirmation question deliberately.

7 Download the first practical from GitHub

From your SSH session, return to your server home directory and clone the repository:

```
cd ~
git clone https://github.com/EnvModelling/approximate-methods-practical
```

Then type:

```
ls
cd approximate-methods-practical
ls
```

Try tab completion as well. Type:

```
cd approx
```

but **before pressing Enter**, press the Tab key. The shell should complete the directory name for you.

If the repository already exists because you cloned it earlier, do not clone another copy. Enter the existing directory instead.

8 Finish and check what you can now do

When you are finished, leave the remote server cleanly:

```
exit
```

Before moving on to the scientific practical, make sure you can tick each statement:

- ☐ I can tell the difference between a local terminal and an SSH session on the remote server.
- ☐ I can use `pwd` and `ls` to work out where I am.
- ☐ I can edit a file with `nano -l` and save it.
- ☐ I know that `lpwd` tells me where SFTP will save a downloaded file on my own computer.
- ☐ I can find a downloaded file in Explorer or Finder.
- ☐ I understand that CAPITAL words in command patterns are placeholders, not literal text to type.
- ☐ I know that `rm` deletes files and that `rm -i` is the safer form while I am learning.
- ☐ I can clone a practical repository from GitHub and move into its directory.

If something does not work, show a GTA the **exact command** you typed and the **exact error message**. Do not include your password in screenshots.